SOLID WASTE MANAGEMENT (BCV515C)

Prof. SPOORTHI B B. E., M. Tech., (Ph.D.)

ASSISTANT PROFESSOR

DEPARTMENT OF CIVIL ENGINEERING

GMIT, DAVANAGERE

MODULE 3

Storage, collection, and Transportation of waste

Methods of storage, Storage container types and materials, onsite processing. Methods of collection and collection vehicles, Analysis, and design of Hauled and Stationary container systems with case studies. Transfer stations – feasibility and economic analysis.

ONSITE STORAGE

The storage and handling of the waste at or near the points of generation before they are collected and transported to the treatment or disposal site

Factors to be considered

1. Effects of storage on the waste components
2. Type of container used
3. The container location
4. Collection Methods
5. Public health and aesthetics

1. Effects of storage on the waste composition

An important consideration in the onsite storage of wastes are the effects of storage itself on the characteristics of the wastes being stored.

1. Microbiological Decomposition:

Food and other wastes placed in storage containers will start to undergo **putrefaction** and time extended leads to **breeding place for flies** and malodors compounds are develop.

2. Absorption of fluids:

Because of the components of SW having differing initial moisture contents, **re-equilibration** takes place as wastes are stored onsite in containers (ex: paper with food waste). The degree of absorption depends on the **length of time** the waste is **stores until collection**.

3. Contamination of waste components

Perhaps the most serious effect of onsite storage of wastes is the contamination that occurs. Small amounts of motor oils, cleaners and paints will contaminate major waste portion of waste and reduce the value of individual components for recycling.

Primary storage

Low volume waste sources

(ex: Residential dwelling ,small isolated commercial premises)

Containers used are

1) Temporary container

(ex: paper, plastic carrier bags, cardboard box)

2) Comparatively permanent

(used metal bins, plastic bins, plastic buckets)

Secondary storage

For large scale collections

1. Stationary container

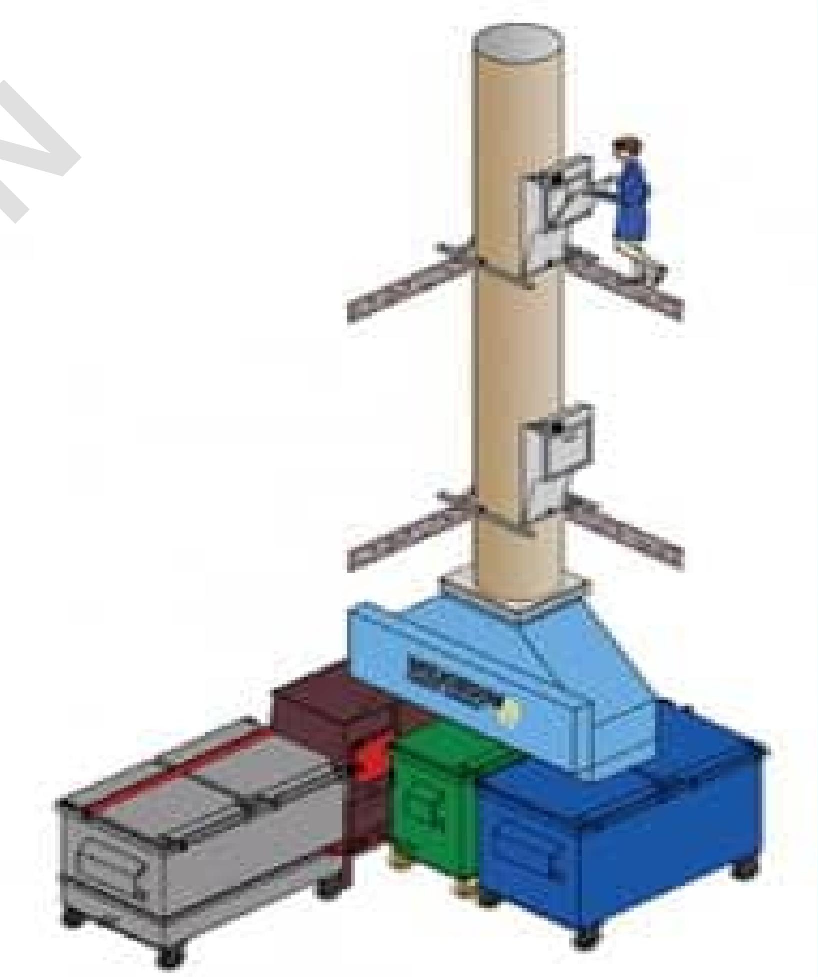
   (open masonry containers, precast ring containers)

2. Portable containers

   (portable container with different capacities)

3. Container Location

Curbside collection



Garbage chute

4. Collection methods

1. DOOR TO DOOR COLLECTION

crew collect from each household in small manually operating vehicle

2. BLOCK COLLECTION

predetermined routes, at intervals of 2 to 3 days, with belly sound

household hand over the container to crew

no container left outside

IMC worker during door to door waste collection

3. CURBSIDE COLLECTION

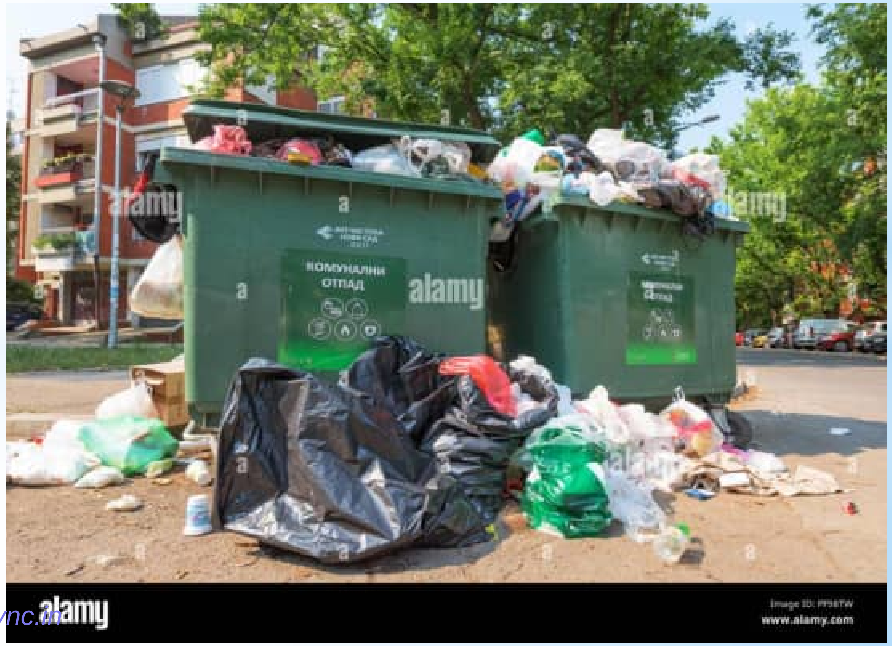
Household place the container at curbside

Crew empty and place back container

Household responsibility to take back container

4. COMMUNAL COLLECTION

Household discarded waste at communal storage facility



5. Public health and aesthetic

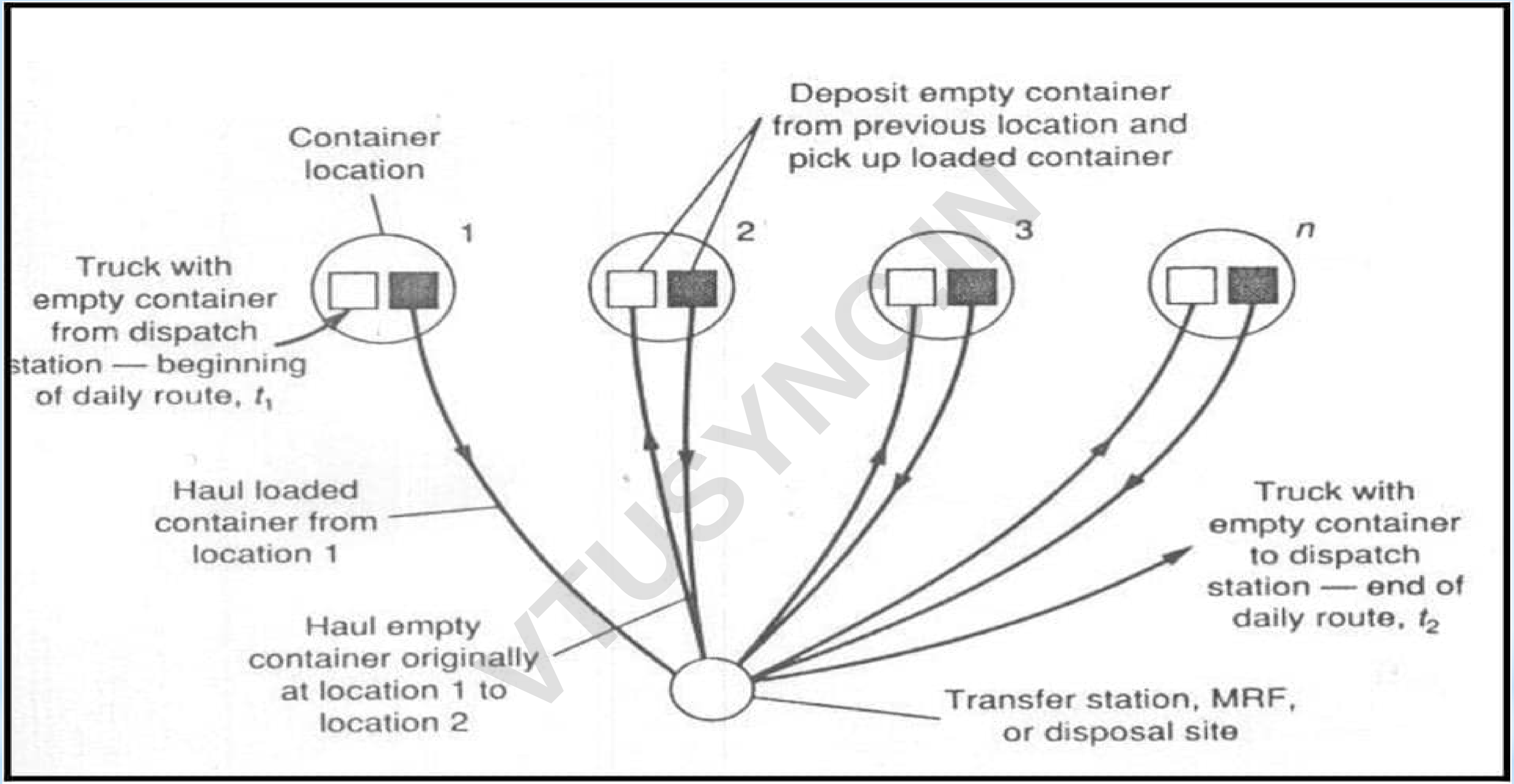
Protection for diseases causing vectors, Odour nuisance and unsightly condition.

- Use tight lids for collection
- Periodic washing of container and storage area
- Periodical removal biodegradable

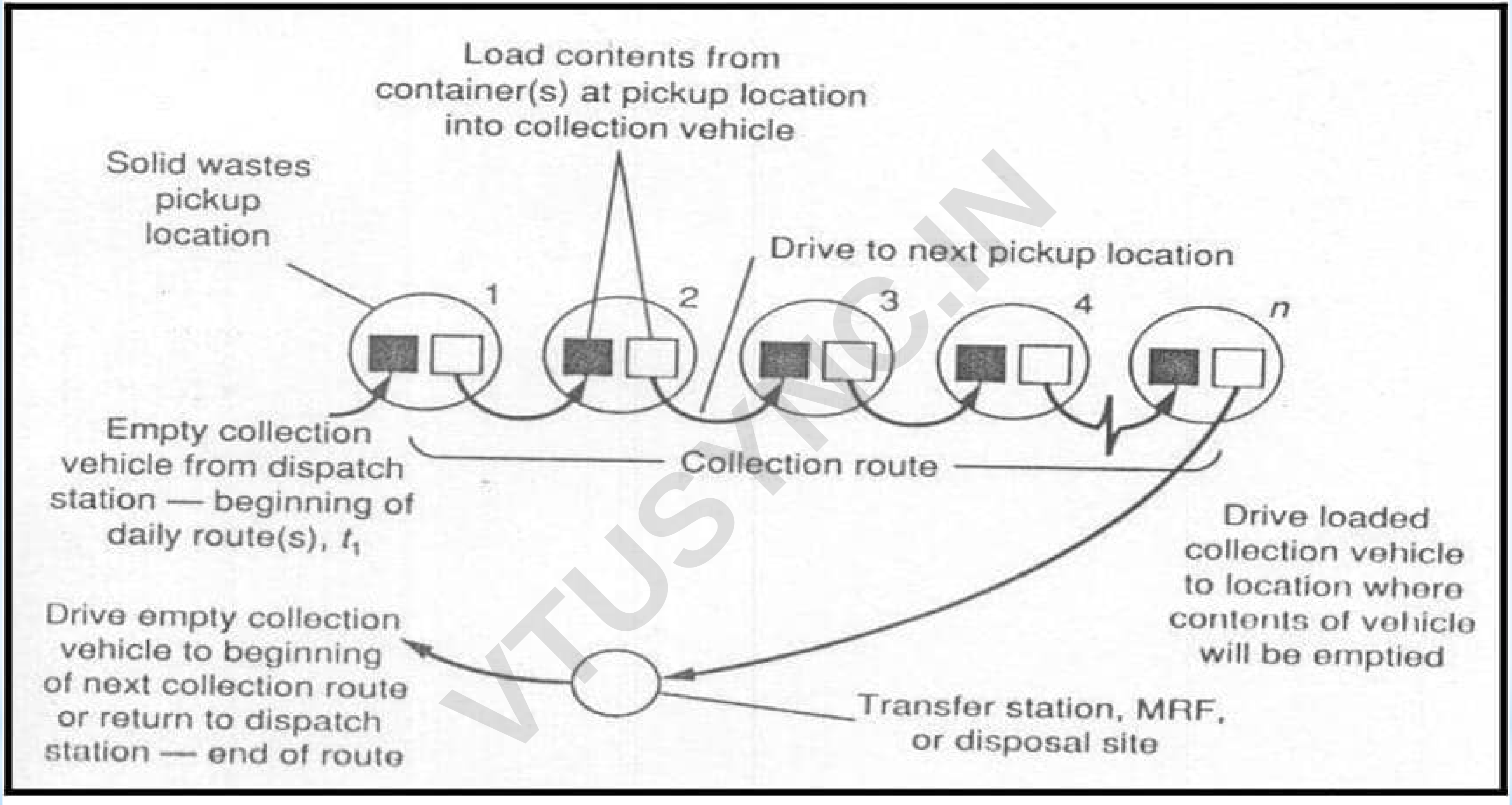
Types of collection services

1. **Residential collection services**
   - Curb services
   - Alley services
   - Set out and set back services
   - Set out services
   - Backyard carry services
2. **Commercial collection services**
   - by Compactions or by Bales

COLLECTION SYSTEMS



HAULED COLLECTION SYSTEM



STATIONARY COLLECTION SYSTEM

TECHNICAL TERMS IN COLLECTION SYSTEM

1. Pick up Time
2. Hauling Time
3. At Site Time
4. Off Route Time

Table (3.1) Definition of terms for the activities involved in the collection of solid wastes.

Term	Definition
Pick up (p)	
Hauled–container system P_{hcs}	The time spent picking up the loaded container, the time required to redeposit the container after its contents have been emptied, and the time spent driving to the next container.
Stationary–container system P_{scs}	The time spent loading the collection vehicle beginning with the stopping of the vehicle point to load the content of the first container and ending when the contents of the last container to be emptied have been loaded.
Haul (h)	
Hauled–container system h_{hcs}	The time required to reach the disposal site, starting after the container whose contents are to be emptied has been loaded on the truck , plus the time after leaving the disposal site until the truck arrives at the location where the empty container is to be redeposited. Time spent at the disposal site is not included.
Stationary–container system h_{scs}	The time required to reach the disposal site, starting after the last container on the route has been emptied or the collection vehicle is filled , plus the time after leaving the disposal site until the truck arrives at the location of the first container to be emptied on the next collection route. Time spent at the disposal site is not included.
At-site (s)	
	The time spent at the disposal site including the time spent waiting to unload as well as the time spent unloading
Off-route (w)	All time spent on activities that are nonproductive from the point of view of the overall collection operation. Necessary off-route time includes (1) Time spent checking in and out in the morning and at the end of the day (2) Time lost due to unavoidable congestion (3) Time spent on equipment repairs and maintenance. Unnecessary off-rout time includes time spent for lunch in excess of the stated lunch periods and time spent on taking unauthorized coffee breaks, talking to friends, etc.

TRANSFER STATION

- To accomplish transfer of solid waste from the collection and other smaller vehicles to larger equipment's.
- Depending upon the method of loading the transport vehicles there the different transfer stations.

NEED FOR TRANSFER STATION/ OPRTATIONS

- Longer haul distance
- Remote locations and cannot reach by highways
- Integral part of MRF's
- Illegal dumping due to excess haul distance
- Use of small collection vehicle
- Use of hauled containers
- Use of hydraulic or pneumatic collection system

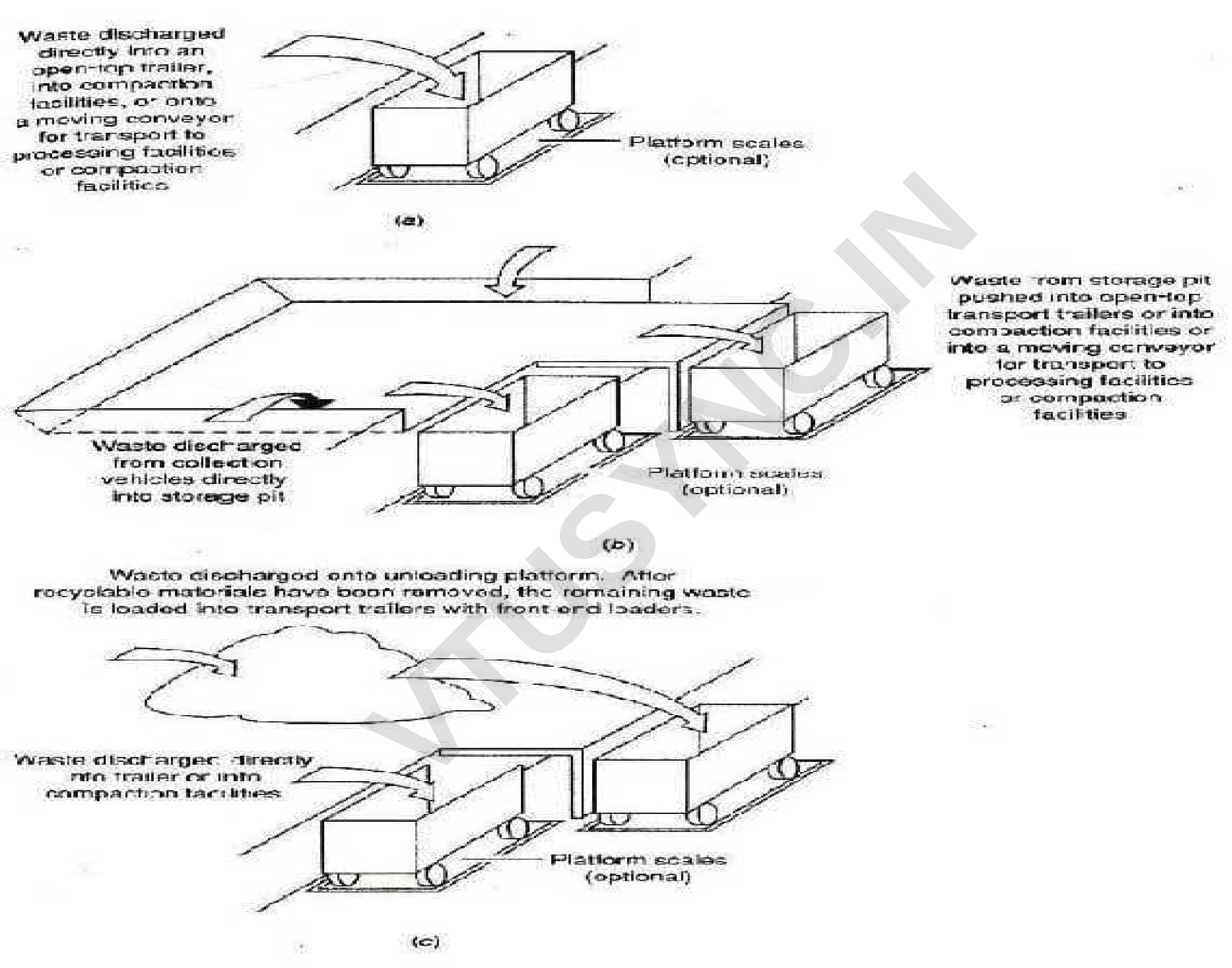
FACTORS INFLUENCING NEED FOR TRANSFER OPERATIONS

- The occurrences of illegal dumping due to excessive haul distance.

- The use of small capacity collection vehicle.

TYPES OF TRANSFER STATIONS

1. DIRECT LOAD
2. STORAGE LOAD
3. COMBINED DIRECT AND STORAGE



Waste discharged directly into an open-top trailer, into compaction facilities, or onto a moving conveyor for transport to processing facilities or compaction facilities
Platform scales (optional)
(a)
Waste from storage pit pushed into open-top transport trailers or into compaction facilities or into a moving conveyor for transport to processing facilities or compaction facilities
Waste discharged from collection vehicles directly into storage pit
Platform scales (optional)
(b)
Waste discharged onto unloading platform. After recyclable materials have been removed, the remaining waste is loaded into transport trailers with front end loaders.
Waste discharged directly into trailer or into compaction facilities
Platform scales (optional)
(c)

Direct load transfer stations

Direct load transfer station with compactors

A storage load transfer station in San Francisco, CA, USA

Waste reduction may occur through the design, manufacture and packing of products with the minimum toxic content, minimum volume of materials and with minimum materials of longer useful life.

- Waste reduction may also occur at the household, commercial or industrial facility through selective buying patterns and by the use of recyclable materials.
- Source reduction is not considered an important element at the present time, because it is difficult to estimate the impact of reduction on the total quantity of waste generated. Never the less, source reduction will likely become an importment factor in reducing the quantity of waste generated.

Some of the other ways in which source reduction can be achieved can be follows

a) Use fewer resources
b) Develop rate structures that encourages to produce less waste
c) Decrease unnecessary and excessive packing materials
d) Substitute recyclable products for disposal

2. Public Attitudes and Legislation

Public attitudes: Ultimately significant reduction in the quantity of waste generated when and only if people are willing to change their habits, lifestyle and to reduce burden associated with management of solid waste. A program of continuing education is essential to bring about change in the public attitude.

Legislation: The most important factors affecting the generation of certain type of waste is the existence of local, state and regulation concerning the use of specific materials. Legislation dealing with packing of products is an example.

3. Geographic location and physical factors on generation of Solid waste

Geographic location and physical factors that affect the quantities of waste generation and collection include location season of the year, waste collection frequency and characteristics of the service area.

Geographic location – The influence of geographic location, different climates that influences both the amount of certain types of solid waste generated and the time period over which waste is generated. For example substantial variation in the amount of yard and garden waste generated in various parts of the country are related to climates. That is in warmer southern areas, where the growing season is longer than northern areas, yard wastes are collected not in considerable amounts but also over longer time.

Season of the year: The quantity of certain type of waste are also affectedby the season of the year.

Collection of Solid Waste

Collection of separated or unseparated solid waste in urban areas is difficult and complex because of the generation of residential, commercial or industrial solid waste takes take place in every home, apartment, commercial or industrial facility, streets, parks and even in vacant areas.

- **Types of Collection Service**

The Various types of collection service are

1. Municipal Collection service / Residential collection service
   A. From low rise detached dwellings
   B. From low and medium rise apartment
   C. From high rise apartment
2. Commercial and Industrial collection service

a. Municipal Collection service / Residential collection service

Collection service varies depending upon the type of dwelling unit, collection for low rise detached dwellings and collection for medium and high rise apartments are considered separately.

The most common types of residential services in various parts of the country include

1. Curb
2. Alley
3. Set out – Set back
4. Set out
5. Backyard carrying

1. **Curb** – is used for low rise detached dwelling. This is a manual type of collection system where in the waste are collected in a curb on a collection day and the containers are returned back to their storage location until the next collection.
2. **Alley** - are part of the basic layout of a city or a given residential area. Alleys are storage of container used for solid waste collection.
3. **Set out – Set back** – Containers are set out from the owners property and set back after being emptied by additional crew.
4. **Set out** – this service is essentially the same as set-out & set-back, except that the home owner is responsible for returning the container back their storage location.
5. **Backyard Carrying** – The collection crew is responsible for entering the owners property and removing the waste from their storage location.

2. Commercial and Industrial collection service

- Both manual and mechanical means are used to collect waste from commercial and industrial areas. To avoid traffic congestion during the day time, solid waste from commercial establishment in many cities are collected in the late evening and early morning.
- Where manual collection is used, Wastes are put in plastic bags, cardboard boxes and other disposable containers that are placed at curbs for collection.

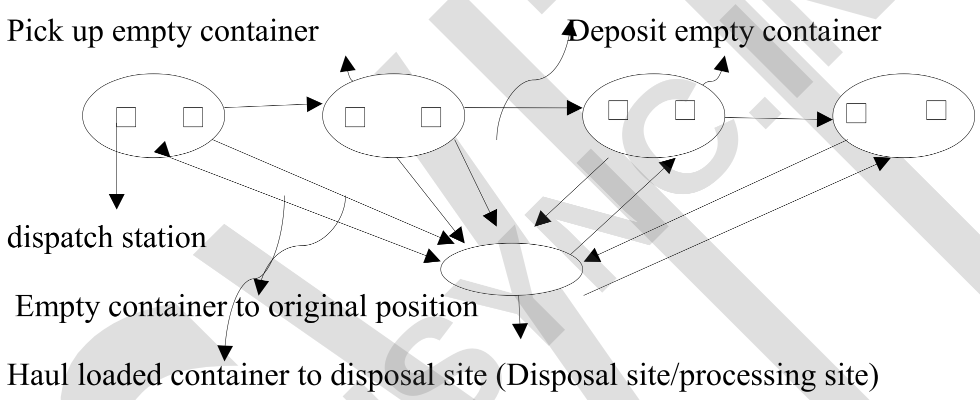
- The Collection services provided to large apartment buildings, residential complexes, commercial and industrial areas are provided to theses centers on the use of movable containers, stationery containers, and large stationery compactors.
- Compactors are of the type that can be used to compress materials into large containers.

❖ **Types of Collection Systems**

Based on the mode of operation, collection systems are classified in to two categories:

1. Hauled container systems
2. Stationery container systems

1. Hauled Container system: Drive to next loaded container



Collection system in which the containers used for the storage of waste are hauled to the processing, transfer or disposal site, emptied and returned to either their original location or some other location are defined as hauled container system. The collector is responsible for driving the vehicle, loading the containers and emptying the contents of containers at disposal site.

There are two main types of hauled container systems

1. Tilt Frame container
2. Trash-trailors

Systems that use tilt frame loaded vehicles often called drop boxes are ideally suited for collection of all types of solid waste and rubbish from locations where generation rates warrants the use of large containers. Because of large volume that can be hauled, the use of tilt frame hauled container systems has become widespread, especially among private services.

Trash trailers are better for collection of especially heavy rubbish such as sand, timber and metal scrap and often are used for collection of demolition waste at construction sites.

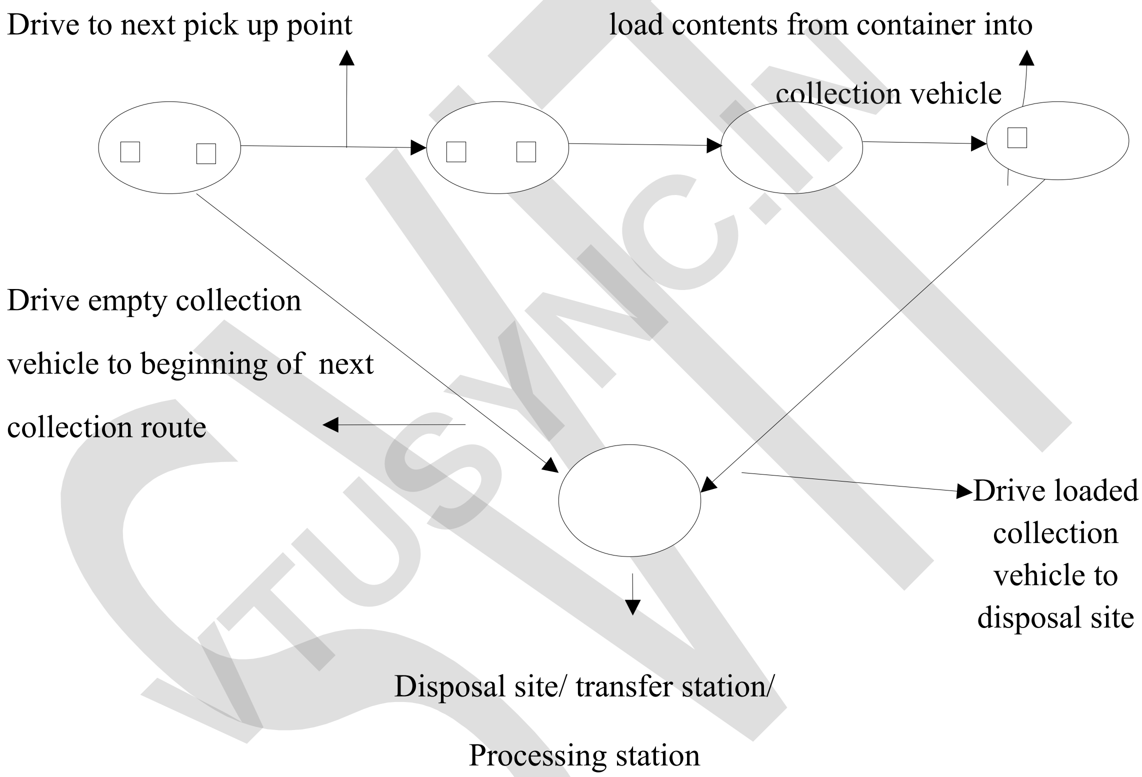
Advantages

1. Hauled container systems are flexible containers of many different shapes and sizes are available for collection of all types of wastes.

Disadvantages

1. Because the container used in this system must be filled, usually the use of very large containers often lead to low volume utilization.
2. Through this system requires only one truck and driver to accomplish the collection cycle, each container picked up requires a round trip to disposal site.

2. Stationery Collection system



Collection system in which the container used for the storage of waste remains at the point of waste generation, except when moved for collection are defined as stationery container system.

There are two main types of stationery container systems

1. Those in which self loading compactors used
2. Those in which manually loaded vehicles are used.

Trips to the disposal site, transfer station, or processing station are made after the contents of a no of containers have been collected and compacted and collection vehicle is full. Because a variety of container sizes and types are available, these systems may be used for collection of all types of wastes.

❖ **Transfer Means and Methods (Transport Methods)**

Motor vehicles, rail road's and ocean going vessels are the principle means used to transport waste. Pneumatic and hydraulic systems have also been used.

1. Motor Vehicle Transport - Motor vehicles used to transport solid wastes on highway should satisfy the following requirement.

- The vehicles must transport wastes at minimum costs
- Wastes must be covered during haul operation
- Vehicles must be designed for highway traffic
- Vehicle capacity must be such that allowable weight limits are not exceeded
- Methods used for unloading must be simple and dependable.

In the recent years because of their simplicity and dependability, open top trailers and semitrailers have found wide acceptance for the transport of waste. Some trailers are equipped with sumps to collect any liquid that accumulate from the solid waste.

2.Rail Road Transport - Although Rail road's were commonly used for the transport of solid waste in the past, they are now used by a few communities. However renewed interest is again developing in the use of rail road for hauling solid waste especially to remote areas where highway travel is difficult and rail lines now exist.

3. **Water Transport** – Barges, scows and special boats have been used in the past to transport solid wastes to processing location and to seasides and ocean disposal sites, but ocean disposal is no longer practiced by developing countries.

4. Pneumatic Transport – Both low pressure air and vacuum conduit transport systems are used to transport solid waste.

- The most common application is the transport of waste from high density apartment or commercial activities to central location for processing or for loading into transport vehicles.

Transfer Station

Important factors that must be considered in the design of transfer station include:

1. Type of transfer operation
2. Capacity Requirement
3. Equipment and accessory requirement
4. Environmental requirement

1. Type of transfer operation : Depending on the method to load the transport vehicles, transfer stations may be classified into three types:

A. Direct Discharge
B. Storage Discharge

C. Combined direct and storage discharge.

A. Direct discharge – In a direct discharge transfer station, wastes from the collection vehicles usually are emptied directly into the vehicle to be used to transport them to a place of final disposal.

- To accomplish this, these stations are constructed in a two level arrangement. The unloading platform from which the wastes from collection vehicles are discharged into the transport trailers are elevated. Direct discharge transfer stations employing stationery compactors are popular.

B. Storage discharge – In this transfer station, wastes are emptied either into a storage pit or onto a platform from which they are loaded into transfer vehicles by various types of auxiliary equipment.

- In a storage discharge transfer station, the various processing techniques like shredding, separation, magnetic separation of ferrous scraps, compaction are employed by using various types of auxiliary equipment.

C. Combined direct and storage discharge – In some transfer station, both the methods are used. Usually these are multipurpose facilities designed to service a broader range of users than a single purpose facility.

2. Capacity Requirement – The operational capacity of a transfer station must be such that the collection vehicles do not have to wait too long to unload.

- In most cases it will not be cost effective to design the station to handle the ultimate peak no of hourly loads. An economic analysis should be made between the annual cost for the time spent by the collection vehicle waiting to unload against the incremental annual cost of a transfer station or the use of more transport equipment.
- Because of the increased cost of transport equipment, trade off analysis must also be made between the capacity of transfer station and the cost of the transport operation including both equipment and labor components.

3. Equipment and Accessory Requirements - The types and amounts of equipment required vary with the capacity of a station and its function in the waste management system.

- Specifically scales should be provided at all medium and large transfer station both to monitor the operation and to develop meaningful management and engineering data.

4. Environmental Requirements – Most of the large transfer stations are enclosed and are constructed of materials that can be maintained and cleaned easily.

- For direct discharge transfer stations with open loading areas, special attention must be given to the problem of blowing papers. Wind screens or other barriers are to be used.
- Regardless of the type of transfer station, the design and construction should be such that all accessible areas where rubbish or paper can accumulate or eliminated.

❖ Factors affecting the location of transfer station

Whenever possible, transfer station should be located

1. As near as possible, the weighted center of the individual solid waste production areas to be served
2. Within easy access of major arterial highway routes as well as near secondary or supplemental means of transportation.
3. Where there will be minimum public and environmental objections to the transfer operations.
4. Where construction and operation will be economical
5. Additionally if the transfer station site is to be used for processing techniques involving materials recovery and energy production, the requirement for those operation must be considered.

❖ Route Optimization

Once the equipment and labor requirements are determined, collection routes must be laid out so both the collectors and equipments are used effectively. In general the layout of collection route involves a series of trial. There is no universal set of route that can be applied for all the situations.

Some guidelines that should be taken into consideration when laying out routes are as follows.

1. Existing policies and regulation related to such items as the point of collection and frequency of collection must be identified.
2. Existing system characteristics such as crew, size and vehicle type must be co-ordinated.
3. Whenever possible route should be laid down so that they begin and end near arterial roads using topographical and physical barriers as route boundaries.
4. In hilly areas, route should start at the top of the grade and proceed downhill as the vehicle is loaded.
5. Routes should be laid out so that the last container to be collected on the route B located nearest to the disposal site.
6. Waste generated at traffic congested location should be collected as early as possible.
7. Sources at which extremely large quantities should be serviced during the first part of the day.

8. Scattered pick up points where small quantities of solid waste generation that receive the same collection frequency should if possible be serviced during one trip or on the same day.

(A)

METHODS OF TRANSPORT

The Various means of transporting the solid wastes are as follows:-

1) ROAD WAYS

* Motor vehicles can be used to transport solid wastes on highways using transporting haulers.
* Simple and easy to operate.
* Widley used in different countries.
* Precautions to be taken in roadways for transportation of solid wastes
  - → Vehicle must transport waste at minimum cost
  - → Waste must be coverd during hauling operation
  - → vehicle loaded should not exceed its limit.
  - → Unloading method should be simple.
  - → slef compacting vehicles should be used
  - → GPS should be used for route optimization.

2) RAILWAY TRANSPORT

* Extensively used in past not nowa days.
* Used where area to be served is having topography of more longitudanal than horizantal (ex: Mumbai)

3) WATERWAYS & NAVIGATIONAL CHANNELS

* solid wastes is transported using barrages, cargo ships and special boats.
* Used where ocean disposal is adopted
* Ocean disposal is not practiced in developed countries.
* Used in developing countries where not much land is available for disposal.
* care should be taken while disposal.

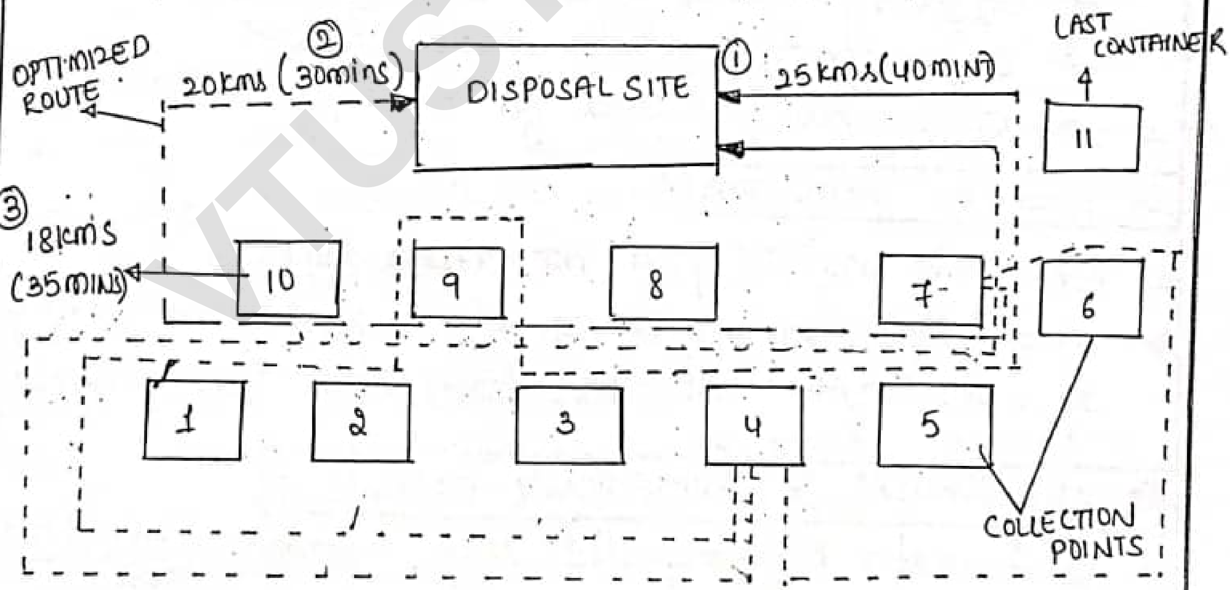
4) PNEUMATIC SYSTEM

* In this method high pressure and low pressure along with vaccume conduit system can be used as means of transport.
* ~~Suitable for high density apartments where waste from~~ individual houses can be brought to central location for processing and loading in to transport vehicle.
* Largest pneumatic system is used in WALT DISNEY ABUSEMENT IN florida, USA.

5) HYDRAULIC TRANSPORT SYSTEM

* Hydraulic force is used to transport waste in industries from different areas to central location for processing ~~or~~ recovery operation.

ROUTE OPTIMISATION



* After deciding labour and transportation equipments, the collection routes must be laid out.
* Layout of collection route involves series of trails & best route is selected which uses labour & equipment effectively

Guides lines to be followed while planning routes are as given below:

1) Existing rules and policies laid down by the authorities related to transportation of solid waste should be followed.
2) Points of collection must be identified
3) Frequency of collection must be identified
4) Vehicle type and creep size must be decided based on requirement.
5) The begining and ending of planned route must be nearer to road.
6) . In hilly areas route should start at top of grade & procede downly hill as vehicle is loaded.
7) Last container to be collected on the route should be located nearer to disposal site.
8) Waste generated at traffic congestion should be collected as early as possible
9) Sources at which extremly large quantity of waste is generated should be serviced during 1st part of the day
10) Where small quantity of waste is generated should be collected frequently.

STEPS INVOLVED IN PLANNING ROUTES

* Preparation of location maps showing significant data and information concerned to waste generation sources.
* Data analysis and preparation of information summary table
* Preliminary lay out of route should be planned and trail runs should be made.
* Evaluation of preliminary routes & development of balenced routes by succesive trails.

Dr. Kiran Kumar B.S.
B.E.,M.Tech(Env.Engg.), Ph.D, MISTE
Assistant Professor
Dept. of Civil Engineering
G.M.I.T. DAVANAGERE.
Mob: 9663369481

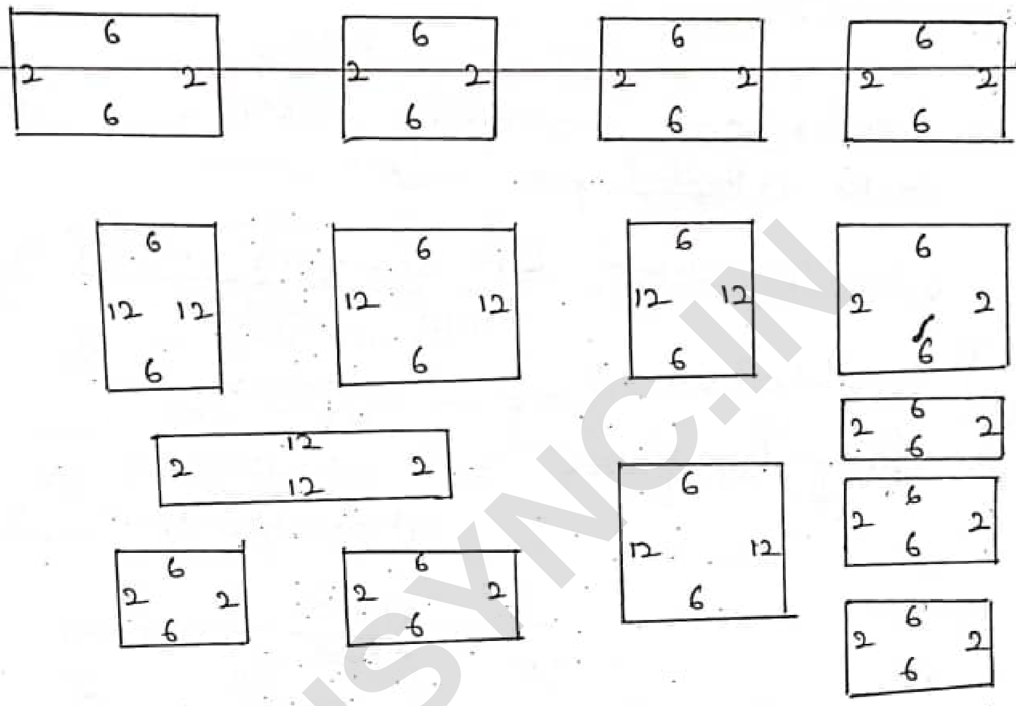
* optimum route must be selected based on data analysis and trail routes.
* Routes are planned using GPS facility, a reciever is placed on container which sends signal to transportation creep when container is full.
* Effectively using GPS technology with available resources

LOCATION OF TRANSFER STATIONS

* should be located as near as possible to the areas to be served
* should be nearer to the highway routes and secondary means of transportation.
* should be located where there is a minimum of public and environmental objection to transfer operation.
* Located where construction & operation is economical.
* options must be given for processing operations
* options must be given for materials recovery & energy production.

* Reliable to complete designated work
* Safety to operation
* Environmental effect
* Earth hazordus
* Asthetics

ROUTE OPTIMIZATION PROBLEMS



The layout of collection routes for residential area is shown in a figure.

Assume the following data applicable and optimise route for collection of muncipal waste

GENERAL: a) occupants per residence = 3.5

b) Solid waste generation rate = 1.6 kg/person/day

c) Type of collection service = curb service

d) collection crew size = 1 person

e) collection vehicle capacity = $20 m^3$

f) compacted density of solid waste in collection vehicle = 325 kg/m3

ROUTE CONSTRAINTS:

a) NO - U turn in streets

b) collection from each side of street with right hand right collection

Dr. Kiran Kumar H.S.
B.E.,M.Tech(Env.Engg.)., Ph.D., MISTE
Assistant Professor
Dept. of Civil Engineering
G.M.I.T. DAVANAGERE.
Mob: 9663368484.

Soln: Determination of total no of residences from which wastes are to be collected.

1) Residences $= (36 \times 4) + (16 \times 10) + (28 \times 1)$

$= \underline{332}$

2) Determine compacted volume of solid waste to be collected per week

Volume per week $= \dfrac{332 \times 3.5 \times 1.6 \times 7}{325}$ kg/m3

$= \underline{40\ kg/m^3}$

3) No of trips / Week $= \dfrac{\text{volume}}{\text{vehicle capacity}}$

$= \dfrac{40m^3}{20m^3}$

$= \underline{2\ \text{Trips}}$

Avg no of residences from which waste are to be collected each day = Residences per trip

$= \dfrac{332}{2}$

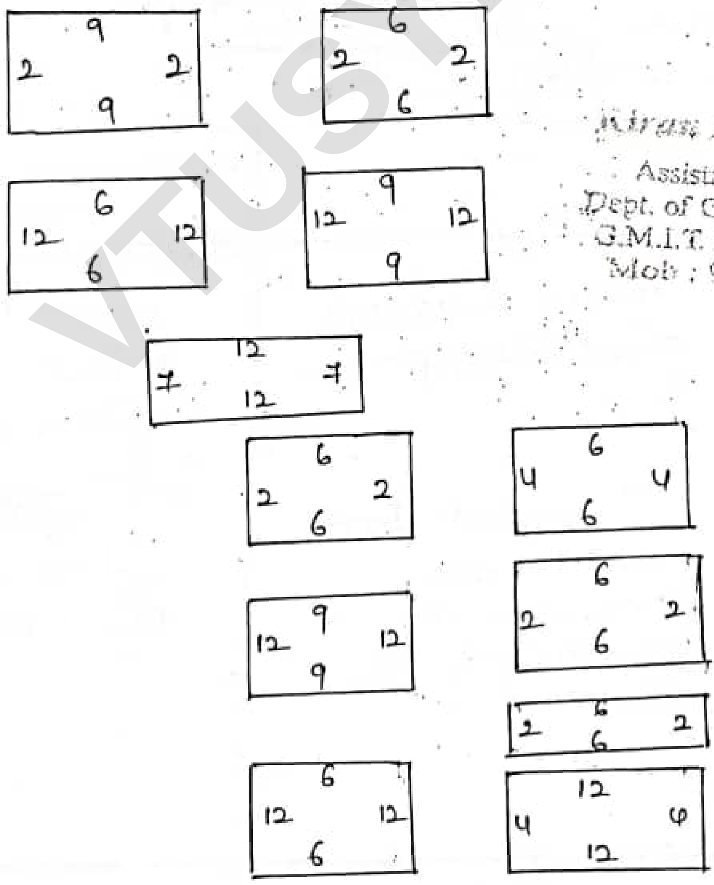
$= \underline{166}$

Kiran Kumar H.S. B.E., M.Tech.
Assistant Professor
Dept. of Civil Engineering
G.M.I.T. DAVANGERE
Mob : 9663368484

2) optimise route for given residential layout in the following figure assume data as follows :

1) occupants per residence = 4
2) Solid waste generation rate = 1.4 kg/person/day
3) Type of collection service = curb service.
4) collection creep size = 1 person
5) collection vehicle capacity = $25 m^3$
6) compacted density of solid waste in collection vehicle = $328 kg/m^3$

Also assume that no route constraints and no U-turns in streets, with stand up right hand right collection vehicle.



Residences $= 22 + (16\times4) + (36\times2) + (42\times2) + 38 + 20 + 32$

$= \underline{\underline{332}}$

Volume / Week $= \dfrac{332\times4\times1.4\times7}{328}$

$= \underline{\underline{39.678}}$ kg/m³

No of trips $= \dfrac{\text{Volume}}{\text{vehiclecap}}$

$= \dfrac{40}{25}$

$= 1.6 \simeq 2$ trips

COLLECTION EQUIPMENTS

TYPICAL DATA ON VEHICLES AND CONTAINERS WITH VARIOUS COLLECTION SYSTEMS

VEHICLE	COLLECTION CONTAINER TYPE	TYPICAL RANGE OF CONTAINER CAPACITIES m³	UNLO METH
Hauled container system			
Tilt frame	- open top, also called as drop boxes	8-40	Gravity inclined tipping
	- used in conjunction with stationary compactor	10-30.	—"—
	- equiped with self contained compaction mechanism	15-30	—"—
Truck-tractor (trash trailors)	- open top trash trailors	10-30	gravity inclined tipping
	- Enclosed trailor mounted containers	15-30	—"—
	- Equiped with self-contained compaction mechanism		
Stationary container systems (mechanically loaded) Front loading Side loading Rear loading	open top & enclosed top & side loading	0.6-8	Hydraul ejector panel
Manually loaded Side loading Rear loading	Small plastic or galvanised metal containers, disposable paper & plastic bags.	75-200	—"—

Dr. Kiran Kumar H.S.
B.E.,M.Tech(Env.Engg.), Ph.D., MISTE
Assistant Professor
Dept. of Civil Engineering
G.M.I.T DAVANAGERE.
Mob: 9663368484

Problem on Hauled-Container Collection System

Solid waste from an industrial park is to be collected in large containers, some of which will be used in conjunction with stationary compactors. Based on traffic studies on similar parks, it is estimated that the average time to drive from the garage to the first container **(t_1)** and from the last container to the garage **(t_2)** each day will be **15** and **30** minutes respectively. If the average time required to drive between containers is **6 minutes** and the one-way distance to the disposal site is **30 km** (speed limit: **88km/h**), determine the number of containers that can be emptied per day, based on a **7-hr working day**.

Solution:

(a) Determine the pick-up time per trip, **PT_{hcs}**

PT_{hcs} = pc + uc + dbc

= 0.5 + (6/60) = **0.6 h/trip**

pc = pick-up time per trip, h/trip

uc = unloading/depositing time for empty container, h/trip

dbc = avg time spent driving between container locations, h/trip

Vehicle	Collection		Pick up loaded container and deposit empty container, h/trip	Empty contents of loaded container, h/container	At-site time q, h/trip
	Loading method	Compaction ratio, r			
			pc + uc		q
Hauled container (Tilt-frame)	Mechanical	2.0–4.0	0.50		0.129
Stationary container (Compactor)	Mechanical	2.0–4.0		0.050	0.15

(b) Determine the time required per trip, T_{hcs}

T_{hcs} = $PT_{hcs} + q + m + nx$

= 0.6 + 0.129 + 0.016 + 0.011 × 30 × 2

= **1.40 h/trip**

q = at-site time per trip, h/trip

m = empirical haul constant, h/trip

n = empirical haul constant, h/km

x = round trip haul distance, km/trip

Type of haul	Speed limit km/h	m h/trip	n h/km
Communal	88	0.016	0.011
Block	72	0.022	0.014
Kerbside	56	0.034	0.018
Door-to-door	40	0.050	0.025

Adapted from: Peavy et al., 1985

(c) Determine the number of trips per day, M_d

$$M_d = \frac{(1-W)L - (t_1 + t_2)}{T_{hcs}}$$

= [(1-0.17) × 7 – (0.25+ 0.5)]/ 1.40

= 3.11 trips/d

M_d (actual) = 3 trips/d

W = off-route factor = 0.17 (assumed)

L = length of working day, h/d = 7 h

t_1 = time from garage to first container location, h = 15/60 = 0.25 h

t_2 = time from last container location to garage, h = 30/60 = 0.5 h

(c) Determine the actual length of the working day 3 = [(1-0.17) × L – (0.25+ 0.5)]/ 1.40

L = 6.75 h (essentially 7 hrs)

Problem on Stationary-Container Collection System

Solid wastes from commercial area are to be collected using a stationary-container collection system having 5 cubic meter containers. Determine the appropriate truck capacity for the following conditions:

- Container utilization factor = 0.70
- Average number of containers at each location = 2
- Collection vehicle compaction ratio = 2.5
- Container unloading time = 0.15h/container (uc)
- Average drive time between container location = 0.15h
- One way haul distance = 25km
- Speed limit = 88 km/h
- Time from garage to first container location = 0.35h
- Time from last container location to garage = 0.25h
- Number of trips to disposal site per day = 2 (M_{dc})
- Length of working day = 8h

Solution:

(a) Determine the time available for each trip, T_{SCS}

$L = [(t_1 + t_2) + M_{dc} (T_{scs})]/(1- W)$

$T_{scs} = [\{8\times (1-0.15)\} - (0.35+0.25)]/ 2$

Therefore, $T_{scs} = 3.1\text{h}$

L = 8h
W = 0.15 (assumed)
t_1 = 0.35 h
t_2 = 0.25 h
M_{dc} = 2

(b) Determine the pick-up time per trip, PT_{scs}

$T_{SCS} = PT_{SCS} + q + m + nx$

$PT_{SCS} = 3.1 - [0.15 + 0.016 + 0.011\times 25\times 2]$

= **2.38 h/trip**

q = 0.15
m = 0.016
m = 0.011
x = 25 km

(c) Determine the number of containers emptied per trip, C_t

$PT_{scs} = C_t\, uc + (S-1)\, dbc$

$2.38 = C_t \times 0.15 + (0.5C_t - 1) \times 0.15$

$C_t = 11.24 =$ **11 containers/trip**

PT_{scs} = 2.38 h/trip

uc = Container unloading time = 0.15

dbc = 0.15 h

C_t = number of containers emptied per trip, container/trip

S = number of container pick-up locations per trip, locations/trip

= C_t /2 (as 2 containers/location) [1

location = 2 containers Therefore, S

locations = 2S containers So, C_t = 2S ,

i.e. S = C_t /2]

(c) Determine the required capacity of the collection vehicle, V_v

$C_t = (V_v \times z)/ (V_c \times f)$

C_t = 11 Containers/trip

z = compaction ratio = 2.5

V_c = container volume, m^3/container = 5 m^3 **f**

= weighted container utilization factor = 0.7

V_v = 15.4 m^3

16 m^3 or nearest standard size can be used.